

Sparse Beamforming for Massive MIMO Systems. This paper studies sparse beamforming techniques for massive MIMO downlink transmission. We propose a novel algorithm that reduces computational complexity by seventy percent. Our method combines compressed sensing with convex relaxation to recover sparse beamforming vectors. Simulation results show a three decibel improvement in spectral efficiency compared to baseline methods. The proposed approach is validated on a 64-antenna base station configuration.

Related Work. Prior approaches to beamforming rely on exhaustive search which does not scale. Zhang et al. proposed a greedy pursuit algorithm but it suffers from local optima. Our work differs by using an alternating direction method of multipliers solver. This yields a provably convergent algorithm with linear convergence rate under mild conditions.